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Portfolio Optimization: Asset Universe Selection Based on Investor Profiles

Bachelor Thesis

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This is to certify that:

- (i) the thesis comprises only my original work toward the Bachelor Degree
- (ii) due acknowledgement has been made in the text to all other material used

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Acknowledgments

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Abstract

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Chapter 1

Introduction

1.1 Motivation

Portfolio construction is often presented as an optimization problem over a set of available assets. Classical portfolio theory explains how capital can be distributed once that set is known, but it often leaves an earlier question outside the model: which assets should be considered in the first place? This question matters because the same candidate universe is not suitable for every investor. Before portfolio weights are assigned, the asset universe should be checked against the investor’s objectives, risk tolerance, and the current market conditions.

This issue becomes clearer in markets where asset behavior changes over time. In the Egyptian financial market, risk can shift because of inflation, exchange-rate movements, monetary-policy changes, and market stress. In such conditions, treating an asset as permanently conservative or permanently aggressive can be misleading. A portfolio process that starts directly with final allocation may therefore depend on a candidate universe that is already weak for the target investor profile.

For that reason, this work treats asset-universe selection as a separate decision stage before allocation. The goal is not to compute final portfolio weights. Instead, the study examines how artificial intelligence can support the selection of an investor-suitable universe before an allocation model is applied. Investor suitability is interpreted through three risk-tolerance groups. Conservative investors prioritize risk control, balanced investors seek a risk-return tradeoff, and aggressive investors accept higher risk in exchange for higher return opportunity.

The implementation studies this problem through an adaptive risk-scoring approach. A reinforcement-learning model predicts an asset-level composite realized-risk score and ranks the active universe in each decision month. The realized-risk target combines realized volatility, downside deviation, and maximum drawdown. These predictions are then used to form risk-tolerance-oriented asset universes before allocation. The contribution is therefore AI-supported asset-risk prediction for investor-suitable universe construction, not final portfolio optimization.

1.2 Problem Statement

Existing portfolio allocation methods commonly assume that the investable universe has already been defined. Related AI and machine-learning preselection studies show that filtering assets before optimization can be studied as its own step. However, these studies commonly select assets using expected return, price movement, profitability, Sharpe-like measures, efficiency, robustness, or final optimizer performance. The problem addressed in this thesis is how to select an investor-suitable asset universe before allocation when suitability is defined by risk tolerance and asset-level realized-risk behavior.

1.3 Scope and Asset Universe

The thesis focuses on a mixed Egyptian-market universe. The investable asset universe includes money-market exposure represented by 91-day Treasury bills, government bonds represented by 5-year bonds, Egyptian equity exposure represented by EGX30 and available EGX30 constituent stocks, real-estate exposure represented by a REIT series, and gold exposure represented by 24K gold in Egyptian pounds. Macroeconomic inputs, including USD exchange-rate behavior and CPI, are used only as model features. They are not treated as investable assets.

The current scope ends at pre-allocation universe selection. Final portfolio weighting, investor-facing production serving, and allocation simulation are not treated as the main implemented contribution. They are later stages that can use the selected universe produced by the proposed framework.

1.4 Research Questions and Objectives

This thesis is guided by the following research questions:

- RQ1: Can AI/ML support dynamic asset-universe selection before allocation using asset-level realized-risk prediction?
- RQ2: Do the selected universes align with conservative, balanced, and aggressive investor risk-tolerance profiles?
- RQ3: How does direct asset-risk prediction differ from existing AI/ML preselection criteria such as return prediction, price movement, Sharpe-like performance, efficiency, or optimizer quality?

The objectives of the thesis are to:

- design a pre-allocation asset-universe selection framework guided by investor risk tolerance;
- construct point-in-time asset-risk features and composite realized-risk targets;
- train and evaluate a reinforcement-learning model that predicts asset-level realized risk;
- evaluate the selected universes against the full active universe using ranking and equal-weight portfolio diagnostics.

1.5 Outline

The remainder of this thesis is organized as follows. Chapter 2 reviews the relevant literature on portfolio optimization, AI/ML asset preselection, investor-risk personalization, reinforcement-learning methods, and the research gap addressed by this work. Later chapters present the methodology, experimental evaluation, and conclusions of the proposed framework.

Chapter 2

Literature Review

2.1 Framing the Problem

This chapter studies asset-universe selection as a stage that comes before final portfolio weighting. An optimizer can only allocate among assets that are included in the candidate set. If that set contains assets that do not match the investor’s risk tolerance, then the allocation stage starts from a weak position. For this reason, the review covers classical portfolio optimization, AI/ML asset preselection, investor-risk personalization, and adaptive reinforcement-learning methods.

The review is organized around one central distinction. Many existing papers show that selecting assets before portfolio optimization can improve later portfolio construction. However, a suitable asset is often defined through expected return, price movement, profitability, Sharpe-like performance, efficiency, robustness, or final optimizer quality. In this work, suitability is defined differently. The focus is whether assets can be selected for conservative, balanced, and aggressive investor groups by directly predicting asset-level realized risk before allocation.

2.2 Portfolio Optimization and the Fixed-Universe Assumption

A classical starting point is the mean-variance framework introduced by Markowitz [1]. The framework formalizes the tradeoff between return and risk when assigning portfolio weights, but it assumes that the investable universe is already known. This assumption is important here because it separates two decisions that are often combined in practice: choosing the candidate assets and assigning final weights.

Later allocation methods provide additional ways to represent investor views and risk objectives. The Black-Litterman model incorporates investor views into allocation

decisions [2], while conditional value-at-risk optimization focuses on downside tail risk [3]. Quantile-based portfolio selection also extends the discussion beyond variance by studying distributional risk criteria [4]. These approaches are relevant because they show that risk can be modeled in several ways, but they still mainly operate after the asset universe has been defined.

Benchmarking work also shows why the universe-selection stage should be separated from the weighting stage. DeMiguel et al. [5] showed that naive diversification is a difficult benchmark for optimized portfolios to beat, while hierarchical risk parity later offered a risk-based allocation alternative that emphasizes diversification structure [6]. These methods are useful baselines for allocation, but they do not decide which assets are suitable for a specific investor before weights are computed.

2.3 AI/ML Asset Preselection Before Optimization

The closest baseline family to this thesis is AI/ML-based asset preselection before portfolio optimization. Wang et al. [7] proposed a deep-learning preselection framework where selected assets are passed to portfolio formation. This paper is directly relevant because it treats asset filtering as a separate stage before allocation. Its selection logic, however, is not based on direct prediction of an asset-level realized-risk target for investor risk-tolerance grouping.

Similar predict-then-select patterns appear in later studies. Ma et al. [8] connected machine-learning and deep-learning return prediction to portfolio optimization. Kaczmarek and Perez [9] used machine-learning predictions to build portfolios under rules such as mean-variance optimization, hierarchical risk parity, and naive diversification. Chaweewanchon and Chaysiri [10] used predictive stock selection before Markowitz optimization. These studies show that ML can serve as a filter before final allocation, but the filtering signal is mainly driven by return or price prediction.

Other work broadens preselection beyond simple return prediction. Mills and Anyomi [11] proposed a two-stage robust portfolio construction approach under uncertainty. Hosseinzadeh et al. [12] used data envelopment analysis for asset preselection before portfolio optimization. Abdi et al. [13] combined LSTM modeling with Sharpe-ratio maximization for prospective asset preselection. Chou and Pham [14] used ensemble learning for stock preselection with multiobjective investment optimization and investor-preference framing.

Together, these papers show that preselection before allocation is a defensible modeling stage. They also form the main baseline family for this thesis. The key difference is the definition of suitability. In most of this literature, a suitable asset is one expected to improve return, price prediction, efficiency, Sharpe-like performance, robustness, or optimizer quality. In this thesis, a suitable asset is one whose predicted realized-risk behavior matches the investor's risk-tolerance group before any allocation model is applied.

2.4 Investor Risk Tolerance and Personalization

The preselection literature motivates the stage studied in this thesis, but investor-specific suitability also requires work on personalization. Personalized finance advisory systems show that recommendations can change when investor information is modeled explicitly. Musto et al. [15] used case-based recommendation and diversification strategies for personalized finance advice. Although the output is not a formal asset universe, the paper supports the idea that financial recommendation should not treat all investors as identical.

Recent AI-based personalization studies provide additional support for investor-aware financial decisions. Asemi et al. [16] used an adaptive neuro-fuzzy inference system to customize investment type from investor demographics and feedback. A related model combined ANFIS, clustering, expert feedback, and multimodal neural-network pretraining for investment-type recommendation [17]. Wei and Liu [18] proposed personalized fund recommendation with dynamic utility learning. These papers connect investor information to financial recommendations, but their outputs are usually investment types or fund recommendations rather than an asset-level universe selected before allocation.

Risk tolerance is also studied directly in robo-advising and personalized portfolio design. Alsabah et al. [19] modeled how investor risk preferences can be learned from portfolio choices. Yu and Liu [20] proposed a personalized mean-CVaR portfolio optimization model for individual investment. Capponi et al. [21] studied personalized robo-advising through client interaction. Schneider and Yilmaz [22] showed that risk appetite can change stock and portfolio recommendations produced by ChatGPT. These studies justify the use of investor risk tolerance as a decision variable, but they do not usually define an explicit pre-allocation asset universe selected through direct asset-risk prediction.

2.5 Adaptive AI and Reinforcement Learning in Finance

Adaptive AI methods are relevant because financial suitability is not static. Deep reinforcement learning has been used for trading and portfolio-management tasks, including financial signal representation and trading decisions [23]. Jiang et al. [24] proposed a deep reinforcement-learning framework for portfolio management, and FinRL later provided environments and tools for automated stock trading and quantitative finance research [25].

Additional DRL portfolio studies show how adaptive policies can be evaluated in financial settings. Lucarelli and Borrotti [26] studied deep Q-learning for cryptocurrency portfolio management. Soleymani and Paquet [27] used online deep reinforcement learning with an autoencoder representation for portfolio optimization. Pinelis and Ruppert

[28] studied machine-learning portfolio allocation, while Rezaei and Nezamabadi-Pour [29] provided a taxonomy and experimental study of deep reinforcement learning in portfolio management.

Risk-aware reinforcement learning is especially relevant to the implementation direction of this thesis. Choudhary et al. [30] trained risk-adjusted deep reinforcement-learning agents using reward functions related to return, Sharpe ratio, and maximum drawdown. This supports the use of RL in risk-aware financial modeling. However, the action in such work is still usually a trading or allocation decision, not the prediction of an asset-level risk score for building a pre-allocation universe.

The closest paper to the present thesis is the investor-specific DRL framework of Orra et al. [31]. Their pipeline uses volatility-guided grouping to form conservative, moderate, and aggressive risk profiles before DRL allocation. It is highly relevant because it connects investor risk profiles, asset grouping, and adaptive learning. The remaining difference is that the grouping is volatility-guided and stock-focused, while this thesis predicts a composite realized-risk target based on realized volatility, downside deviation, and maximum drawdown over a mixed and variable asset universe.

2.6 Synthesis and Research Gap

The reviewed literature leads to three observations. First, classical allocation methods provide strong tools for assigning portfolio weights, but they usually assume that the investable universe has already been chosen. Second, AI/ML preselection studies show that selecting assets before optimization can be evaluated separately from weighting. Third, investor-personalization studies show that risk tolerance should affect financial recommendations. The gap appears between these streams rather than inside only one of them.

The main baseline family for this thesis is AI/ML asset preselection before portfolio optimization. These studies solve a related problem and should be treated as important comparison work. Their limitation is more specific: the selection criterion is usually return prediction, price movement, profitability, Sharpe-like performance, efficiency, robustness, or optimizer quality. Direct prediction of an asset-level risk target is mostly absent. The closest partial exception is Orra et al. [31], which uses volatility for risk-profile grouping, but it does not predict a composite realized-risk target in the same form used here.

This thesis is positioned as an investor-risk-tolerance-oriented extension of the preselection literature. It studies pre-allocation universe construction by predicting and ranking assets according to a composite `realized_risk` target before final allocation. The selected universes are then interpreted according to conservative, balanced, and aggressive investor profiles. This connects asset preselection, investor risk tolerance, and adaptive AI while keeping final portfolio weighting as a later stage.

Chapter 3

Methodology

This chapter defines the methodology used to construct investor-risk-oriented asset universes. The proposed method scores and ranks assets by predicted realized-risk suitability, then maps the ranked universe to conservative, balanced, and aggressive investor profiles.

The core contribution is a pre-allocation universe construction framework. Portfolio diagnostics are used only as secondary checks of whether the selected universes behave coherently under historical simulation.

3.1 Methodological Scope and Objective

The methodology addresses the gap identified in Chapter 2: risk is widely used inside portfolio optimization objectives, constraints, and evaluation metrics, but is less often treated as the main signal for investor-specific asset eligibility before allocation. The objective is to estimate which assets are suitable for conservative, balanced, and aggressive investors by learning a month-level asset risk ranking from point-in-time market information.

The framework makes two methodological choices. First, asset suitability is represented through a constructed realized-risk target rather than through expected return alone. Second, model development uses strict chronological ordering so that every decision month is evaluated from information available before the relevant realized-risk target is observed.

3.2 Proposed Framework Overview

Figure 3.1 summarizes the end-to-end methodology. Historical daily and macroeconomic series are cleaned into a point-in-time monthly panel. The model predicts one bounded risk score per active asset in each decision month, ranks the active universe from low

to high predicted risk, and maps ranked assets into overlapping investor risk buckets. Equal-weight portfolio diagnostics are then computed on the selected universes as an evaluation layer.

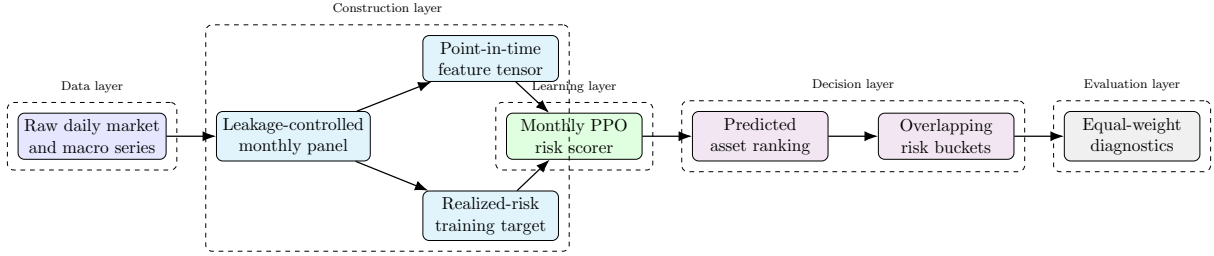


Figure 3.1: Layered methodology pipeline for risk-tolerance-oriented universe construction.

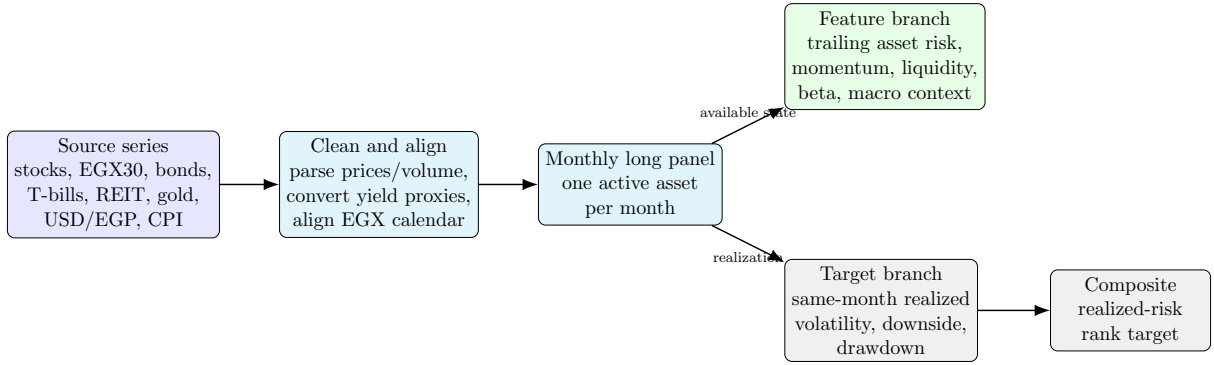


Figure 3.2: Data and target construction from raw source series to monthly feature and realized-risk branches.

The pipeline depends on a constructed risk target rather than a fixed external risk label. The realized-risk target is built by combining realized volatility, downside deviation, and maximum drawdown after cross-sectional normalization. During development, alternative component weights such as equal weighting, 40-40-20, and variants that remove the drawdown component were evaluated before selecting the promoted target profile.

3.3 Composite Realized-Risk Target

The target variable is a composite realized-risk score. It combines realized volatility, downside deviation, and maximum drawdown. These components are chosen because each captures a different form of risk exposure: volatility measures overall return variation, downside deviation focuses on negative variation, and maximum drawdown captures peak-to-trough loss severity.

The Sortino and Calmar ratios express return per unit of risk by changing the denominator: downside deviation for Sortino and maximum drawdown for Calmar. Since this thesis targets asset suitability by risk rather than return maximization, the methodology uses the denominator concepts directly instead of optimizing return-to-risk ratios.

Each component is converted into a within-month cross-sectional rank so that assets are compared against the active universe available in the same month. A larger normalized rank means higher realized risk. The promoted target uses an equal contribution from the three ranked components. Alternative realized-risk profiles were also tested during development, including 33-33-33, 40-40-20, 50-50-0, and other variants that shifted emphasis between volatility, downside deviation, and drawdown. The equal-weight profile is the selected methodology profile.

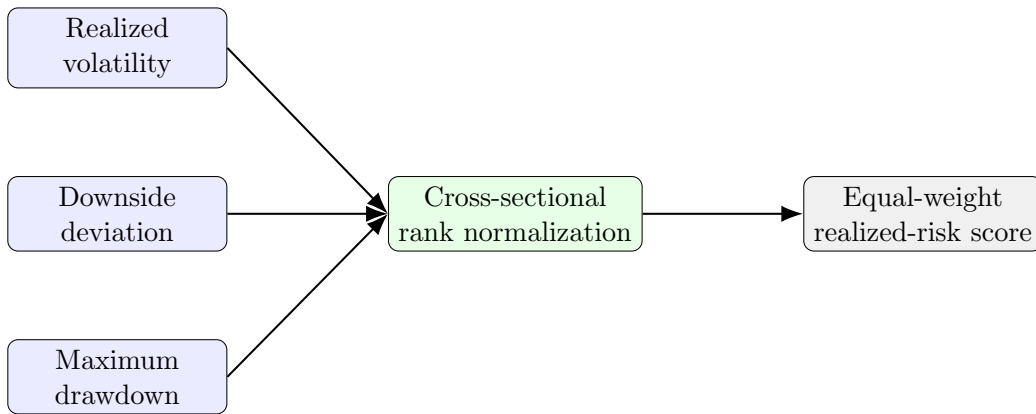


Figure 3.3: Composite realized-risk target construction.

3.4 Dataset and Point-in-Time Feature Construction

The data engineering pipeline converts daily market and macroeconomic data into a monthly long panel. One row represents one active (month, asset) state.

Raw market inputs follow an Investing.com-style CSV format. Files are reverse chronological, use MM/DD/YYYY dates, and contain fields such as price, open, high, low, volume, and quoted percentage change. Numeric fields are parsed after removing formatting characters; volume suffixes such as K, M, and B are converted into numeric values. The vendor change field is retained as a quality-control field, while authoritative returns are computed from cleaned prices. Money-market and bond series are quoted as yields, so fixed-maturity price proxies are used before returns and range-derived features are calculated.

Table 3.1: Raw data sources used by the pipeline

Series	Data type	Methodological role
91-day treasury bills	Yield series	Low-risk money-market exposure; yield converted to price proxy for returns.
5-year government bonds	Yield series	Government fixed-income exposure; yield converted to price proxy for returns.
EGX30 index	Market index	Equity-market benchmark and beta reference.
EGX30 constituents	Daily equity prices	Expanded equity universe; rows begin only when real history exists.
REIT index	Real-estate proxy	Real-estate exposure proxy.
Gold	Commodity price	Gold exposure in Egyptian pounds.
USD exchange-rate series	Foreign-exchange series	Macro/foreign-exchange context after concatenation and deduplication.
CPI	Inflation series	Inflation context; monthly file cleaned separately from daily market data.

Daily series are aligned to the Egyptian market calendar. Short forward filling is applied only to quoted level series where a missing observation reflects non-trading or publication timing rather than a missing asset history. The fill limit is capped at five trading days. Audit fields, range-price fields, volume, and vendor change values remain missing on synthetic forward-filled rows. Pre-listing history is never imputed; an absent row means the asset was unavailable for that month. This rule is central to handling assets with shorter histories.

The feature families used by the model are shown in Table 3.2. Asset-level features are normalized cross-sectionally within each month. Macro features repeat across active assets in the same month and are not cross-sectionally normalized.

Table 3.2: Feature families used for point-in-time asset risk scoring

Feature family	Methodological role
Volatility and risk	Captures recent realized variation, EGARCH volatility summaries, drawdown behavior, and downside variation.
Downside and tail behavior	Captures harmful return asymmetry and recent downside-tail contribution.
Liquidity	Represents trading activity through observed volume where available.
Momentum and technical state	Captures relative price location, range behavior, momentum, and overbought/oversold state.
Market beta	Measures recent co-movement with the EGX30 benchmark.
Macro context	Represents foreign-exchange and inflation conditions shared across the active universe.

Table 3.3: Promoted model feature and target fields

Field	Timing/window	Level	Role
<code>egarch_vol</code>	trailing 3 months	asset	Walk-forward EGARCH volatility summary, aggregated over the trailing 3-month feature window.
<code>downside_dev</code>	trailing 3 months	asset	Annualized downside deviation from recent daily returns.
<code>max_drawdown</code>	trailing 3 months	asset	Recent absolute peak-to-trough drawdown exposure.
<code>volume</code>	trailing 3 months	asset	Sum of observed vendor volume, defaulting to zero when no vendor volume exists.
<code>atr_pct_20</code>	trailing 20 observations	asset	Average true range over the last 20 observations, divided by the last observed close.
<code>beta_to_egx30</code>	trailing 3 months	asset	Daily-return co-movement with the EGX30 benchmark.
<code>price_to_sma20</code>	last state in month	asset	Last observed close relative to the 20-observation simple moving average.
<code>rsi_14</code>	last state in month	asset	Fourteen-observation Wilder RSI state.
<code>distance_to_3m_high</code>	trailing 3 months	asset	Last observed close relative to the recent 3-month high.
<code>usd_vol</code>	trailing 3 months	macro/context	Annualized USD/EGP return volatility, repeated across assets in the month.
<code>cpi_trajectory</code>	trailing 3 monthly observations	macro/context	Compounded CPI month-on-month inflation trajectory, repeated across assets in the month.
<code>downside_tail_ratio_3m</code>	trailing 3 months	asset	Additive feature measuring the share of absolute movement coming from the largest downside-tail losses.
<code>realized_vol</code>	target month	target-only	Within-month realized volatility component.
<code>realized_downside_dev</code>	target month	target-only	Within-month realized downside-deviation component.
<code>realized_max_drawdown</code>	target month	target-only	Within-month realized drawdown component.
<code>realized_risk</code>	target month	target-only	Equal-weight composite of ranked realized volatility, downside deviation, and maximum drawdown.
<code>realized_rank</code>	target month	target-only	Ascending within-month rank of realized_risk , used for ranking evaluation.

Asset identifiers are retained only as metadata. The model-facing tensor excludes `AssetID`, `AssetName`, and `AssetGroup`. This prevents the PPO policy from memorizing ticker identities and forces the shared row scorer to learn market and risk patterns that can generalize across assets. Assets with shorter histories enter the panel only after real observations exist, and the shared scorer can still predict their risk once their point-in-time features are available.

3.5 Why Reinforcement Learning Was Used

The asset-scoring task is not a single static classification problem. Each decision month contains a different active universe, a different market context, and a different cross-sectional risk ordering. The model must therefore score all currently available assets jointly and receive feedback at the month level after the full ranking is formed.

Reinforcement learning was selected because it can optimize the scoring policy directly against a month-level ranking reward rather than treating every asset row as an

independent supervised sample. This matches the methodological objective: the model is judged by how well its predicted ordering matches the realized-risk ordering of the active universe. Supervised tabular models were considered as alternatives, but the promoted design uses a policy-learning formulation with active-universe masking and rank-based feedback.

3.6 Reinforcement-Learning-Based Risk Scoring

The active implementation uses Proximal Policy Optimization (PPO). One PPO episode corresponds to one decision month. The framework acts as a monthly risk scorer: it receives a padded feature tensor and an active-asset mask, then outputs one score for each active asset. Padded rows are used only for fixed tensor shape and do not contribute to action sampling, log probability, entropy, loss terms, reward, or evaluation metrics.

The action is one bounded continuous risk score per active asset. Assets are sorted from low to high predicted score to form the predicted risk ranking. The reward is computed once per month across the active universe:

$$R_t = 0.7 \cdot \rho_s(\hat{r}_t, r_t) + 0.3 \cdot (1 - \text{MSE}(\hat{s}_t, s_t)), \quad (3.1)$$

where ρ_s is the Spearman rank correlation between predicted and realized ranks, $\hat{s}_t$ is the predicted score vector, and s_t is the realized-risk target vector. Reward variants such as rank-only 100-0, balanced 50-50, and rank-dominant 70-30 were evaluated during development. The promoted reward profile uses 70% rank correlation and 30% score-level error.

3.7 Actor-Critic Architecture

The policy uses a masked actor-critic architecture with a shared asset-row scorer. The actor applies the same row encoder to every active asset row, combines row-level representations with pooled universe context, and outputs one risk-score distribution per asset. The critic uses a mask-aware pooled summary of the active universe and produces a scalar value estimate for the decision month.

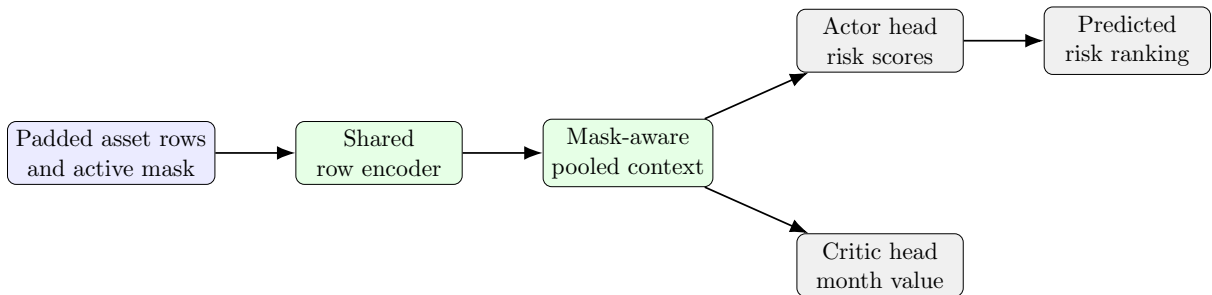


Figure 3.4: Masked PPO actor-critic architecture for variable active universes.

Daily-input variants were evaluated during development. In the daily-flat variant, recent daily observations were zero-padded, masked, flattened, and concatenated into the row scorer. In the daily-strip CNN variant, recent daily observations were arranged as a short temporal strip with channels such as relative close, daily return, transformed volume, and a volume-observed indicator; convolutional filters were then used to summarize local daily patterns before scoring. The promoted thesis model uses monthly feature rows with pooled context rather than a daily-input encoder.

3.8 Method Selection and Alternative Model Families

Random forests and XGBoost were considered as supervised tabular alternatives. They are useful for independent-row prediction, but the promoted thesis method optimizes a month-level ranking policy over a variable active universe with masking and rank-based feedback.

Transformer and attention-first models were considered as sequence or cross-asset attention alternatives. They were not promoted because the implemented development path focused on monthly PPO scoring, pooled context, and daily-input CNN/flat comparisons. Attention remains a possible extension, but no attention model is claimed as part of the implemented methodology.

No NLP encoder is used in the asset-risk scorer because the model inputs are numerical market and macroeconomic series. The web-application questionnaire contains textual option labels, but those labels are converted through deterministic categorical mappings into numeric tabular features; this is not a news/text NLP pipeline and does not feed the PPO asset-risk scorer.

Table 3.4: Compared model families and thesis-safe methodological decision

Model family	Role considered	Methodological decision
PPO actor-critic	Sequential month-level risk ranking over variable active universes	Promoted asset-risk scorer.
Random forest / XG-Boost	Supervised tabular prediction or classification	Considered as supervised alternatives, but not used as the main asset-risk scorer.
Daily-flat encoder	Flattened recent daily paths added to the scorer	Evaluated as a daily-input candidate.
Daily-strip CNN encoder	Convolutional encoding of short daily-return strips	Evaluated as a daily-input candidate.
Transformers / attention	Sequence or cross-asset attention modeling	Considered as a future extension, not part of the implemented method.
Questionnaire categorical encoding	Textual option labels mapped to numeric fields in the application	Used only for application-side investor-profile input handling, not as NLP for asset-risk scoring.

3.9 Investor Risk Profile Mapping and Bucketing

After PPO inference, active assets are ranked from low to high predicted risk. The ranked universe is then mapped to investor profiles using overlapping rank-percentile bands. These percentiles describe the share of assets after sorting, not the raw predicted-risk score. Several bucket methods were evaluated, including non-overlapping terciles, broad 40% tail overlaps, wider 50% overlaps, and the promoted `tail_30_overlap` method. In the promoted method, low risk uses the 0.00–0.30 asset-rank percentile range, medium risk uses 0.20–0.80, and high risk uses 0.70–1.00.

Table 3.5: Investor profile bucket definitions

Investor profile	Asset-rank percentile band	Interpretation
Conservative	0.00–0.30	Selective low-risk universe.
Balanced	0.20–0.80	Broad middle universe with shared boundary assets.
Aggressive	0.70–1.00	Selective high-risk universe.

The overlap is intentional. In practical portfolio management, suitability is not always mutually exclusive. A relatively stable asset can be acceptable for both conservative and balanced investors, while an upper-middle-risk asset can be acceptable for balanced and aggressive investors.

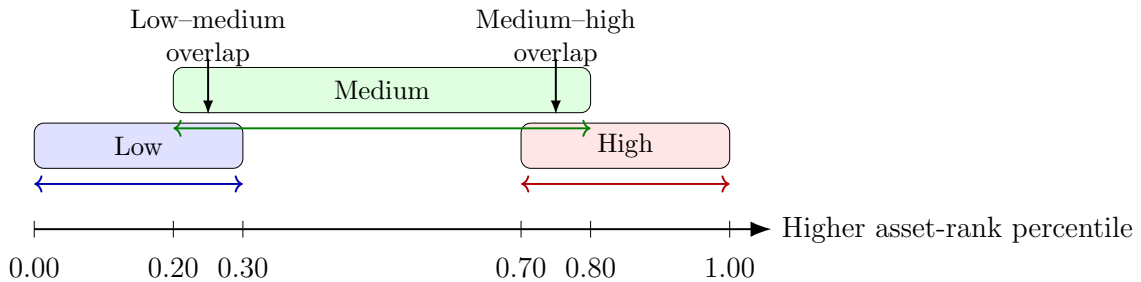


Figure 3.5: Overlapping investor risk buckets on the predicted-risk asset ranking.

3.10 Training and Experimental Protocol

The development protocol was sequential. The data architecture was fixed first to protect point-in-time construction and leakage control. Framework and input architecture comparisons were then run over monthly features without context, monthly features with pooled context, daily-flat inputs, and daily-return CNN inputs. Feature testing followed

through leave-one-out checks, replacement variants such as SMA, ATR, RSI, and window changes, and additive tail/ratio candidates. PPO hyperparameters were tuned with Optuna after the framework was locked. Final candidate reruns and bucket-method evaluation were performed after model selection.

Table 3.6: Experimental protocol summary

Phase	Purpose
Data architecture	Build the daily reference file and monthly point-in-time panel.
Framework comparison	Compare monthly features without context, monthly features with pooled context, daily-flat inputs, and daily-return CNN inputs.
Feature testing	Run leave-one-out checks, replacement variants, and additive tail/ratio candidates.
PPO tuning	Use Optuna with validation reward and Spearman guardrails.
Final reruns	Confirm selected candidates across seeds and promote the final artifact.
Bucket evaluation	Compare overlapping and non-overlapping percentile bucket methods.

The active panel covers October 2010 to January 2026. Decision-month training and evaluation use chronological splits. Training uses January 2011 to December 2022. Validation uses January 2023 to February 2025. Test reporting uses March 2025 to January 2026. The test period is not used for framework, feature, hyperparameter, or bucket selection.

Table 3.7: Promoted PPO hyperparameter configuration

Hyperparameter	Value
Learning rate	2.49×10^{-4}
<code>n_steps</code>	256
<code>batch_size</code>	256
<code>n_epochs</code>	10
γ	1.0
GAE λ	1.0
Clip range	0.299
Entropy coefficient	0.00235
Value-function coefficient	0.902
Maximum gradient norm	0.3

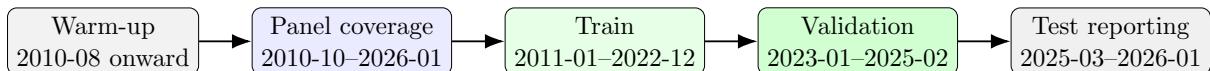


Figure 3.6: Chronological training, validation, and test reporting protocol.

3.11 Evaluation Protocol

Evaluation focuses first on model ranking quality. The primary model check compares the predicted asset ranking against the realized-risk ranking for the same decision month.

Secondary portfolio diagnostics compare an equal-weight full-universe portfolio against equal-weight filtered universes for conservative, balanced, and aggressive profiles. The diagnostic metrics include return, volatility, Sharpe ratio, Sortino ratio, and maximum drawdown.

The evaluation therefore has two layers. The model layer tests whether PPO scores produce a useful realized-risk ordering. The portfolio layer tests whether the selected risk-profile universes behave differently from the full universe under the same equal-weight rule.

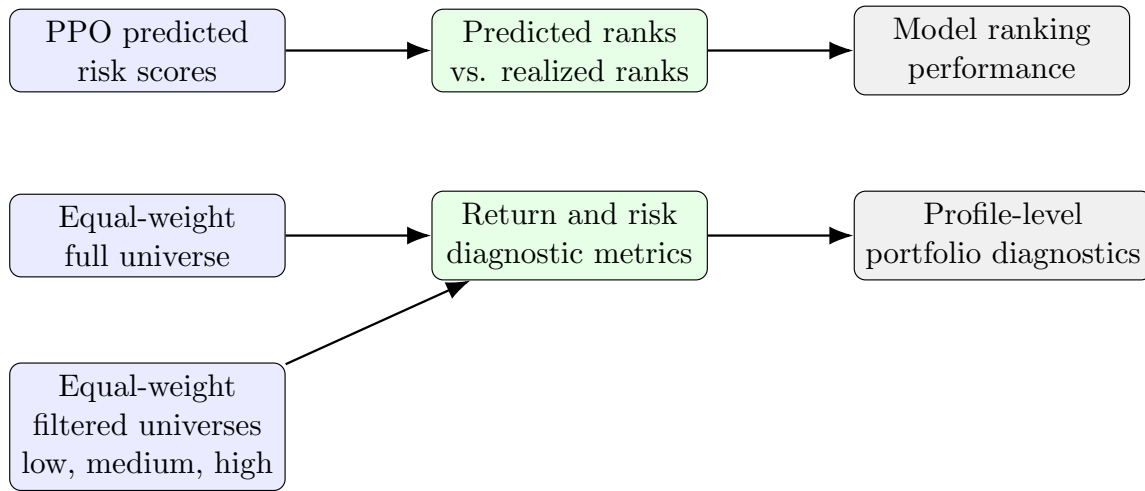


Figure 3.7: Evaluation logic for ranking performance and equal-weight portfolio diagnostics.

3.12 Chapter Summary

This chapter presented a risk-tolerance-oriented methodology for asset-universe construction. Market and macro data are transformed into leakage-controlled monthly asset states; a PPO actor-critic model scores active assets by predicted risk; ranked assets are mapped into overlapping conservative, balanced, and aggressive buckets; and equal-weight diagnostics compare the filtered universes with the full active universe.

Chapter 4

Results & Limitations

4.1 Experiments Setup

Section text.

4.2 Dataset Description

Section text.

4.3 Results

Section text.

4.3.1 Experiment 1

Subsection text.

4.3.2 Experiment 2

Subsection text.

4.4 Results Analysis and Discussion

Section text.

Chapter 5

Conclusion & Future Work

5.1 Conclusion

Section text.

5.2 Future Work

Section text.

Appendix

Appendix A

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